

Circulatory system

Lecture: 18

Circulatory System

□ *Circulatory system in humans, is the combined function of the*

□ *heart,*

□ *blood and*

□ *blood vessels (arteries, veins and capillaries)*

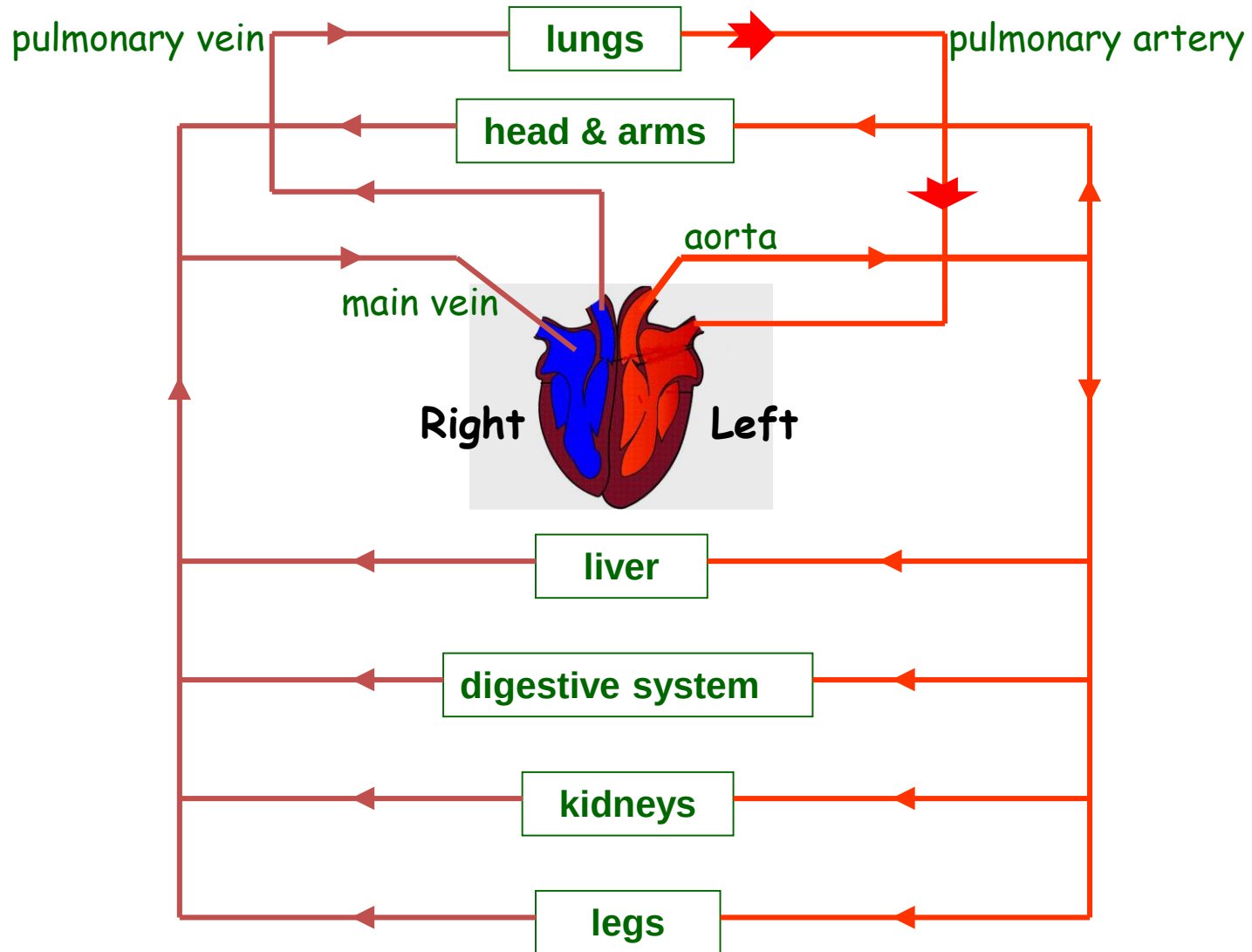
to transport oxygen and nutrients to organs and tissues throughout the body and carry away waste products.

- The arteries, veins, and capillaries in an average adult human and laid them end-to-end, the total length would be **about 96,000–100,000 km**. That's enough to circle the Earth **about 2.5 times**.

Average cardiac output at **rest** = ~5 liters per minute.

- **In 1 hour:** $5 \text{ L/min} \times 60 \text{ min} = 300 \text{ L/hour}$
- **In 24 hours:** $300 \text{ L/hour} \times 24 \text{ h} = \mathbf{7,200 \text{ L/day}}$ ($\approx$ **7–8 thousand liters**)
- During **exercise**, the heart can pump **3–5 times** more than this.

How does this system work?

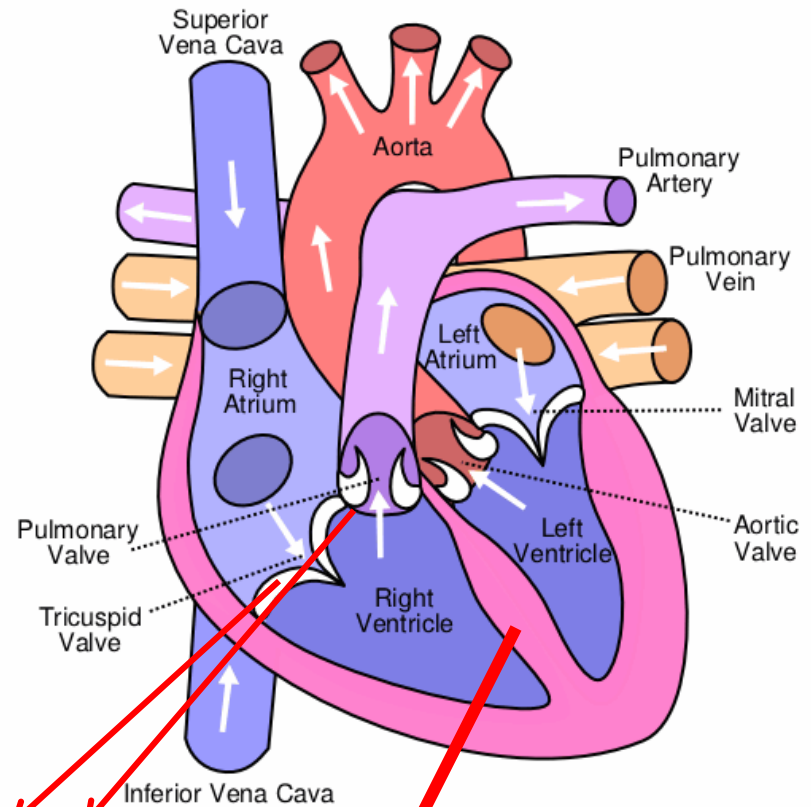


Anatomy- Heart

- The heart provides the pressure needed to keep the blood flowing through the network of tubing.
- The heart is the engine of the circulatory system.
- It is divided **into four chambers**:

- (1) the **right atrium**,
- (2) the **right ventricle**,
- (3) the **left atrium**, and
- (4) the **left ventricle**.

- The two **atria collect the blood**.
- The **two ventricles pump the blood out** of the heart.
- Valves prevent the blood from **flowing backwards**.
- The septum separates the two sides of the heart left and right.
- The **right side** pumps **de-oxygenated blood to the lungs**.
- The **left side** pumps the **oxygenated blood** to the rest of the body.



Valves

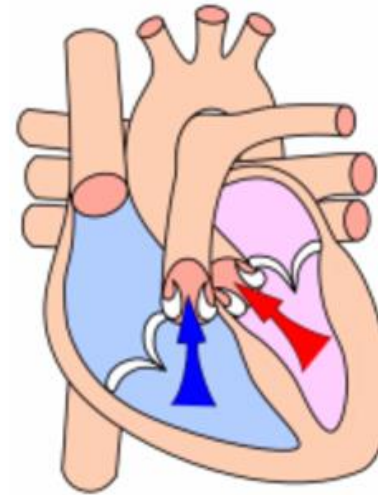
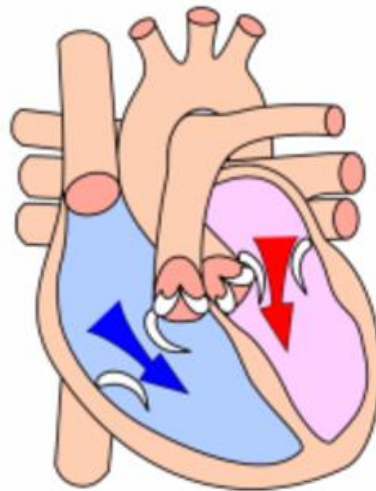
Septum

Heart pump

The pumping action of the heart occurs in two stages for each heartbeat:

- (1) **DIASTOLE**, when the heart is at **rest/dialet** to collect blood; and
- (2) **SYSTOLE**, when the heart **contracts** to pump deoxygenated blood toward the lungs and oxygenated blood to the body.

During each heartbeat, typically about 60 to 90 ml of blood are pumped out of the heart. If the heart stops pumping, death usually occurs within four to five minutes.

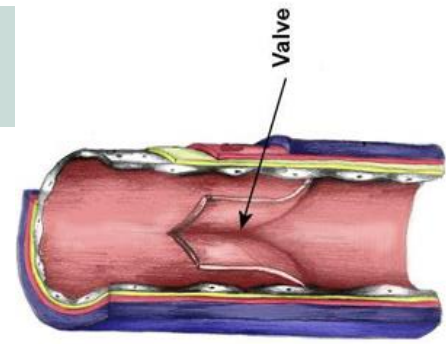
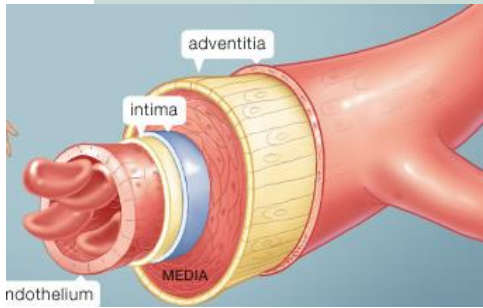


Normal range:

Diastole
60 – 80 mmHg (adults)

Systole
90 – 120 mmHg (adults)

Artery and Vein



Artery

- Carry **oxygenated blood** (except pulmonary artery).
- Carry blood **away from the heart**
- Thick and elastic muscular wall.
- Blood under high pressure.
- Contain **no** valve
- Blood moves in pulses.
- Blood flows rapidly.
- Small arteries are called **arterioles**

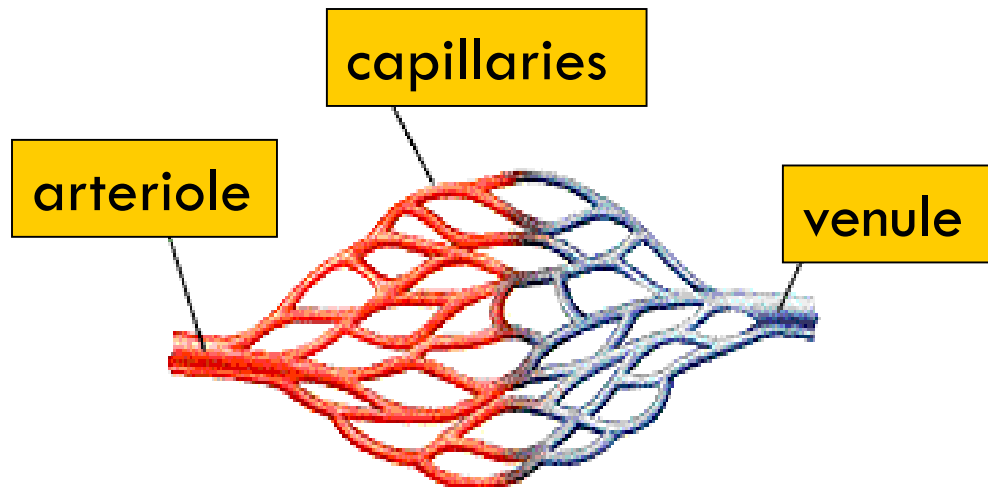
Vein

- Carry **deoxygenated** blood (except pulmonary vein)
- Carry blood **to the heart**
- Thin and slightly muscular wall.
- Blood under low pressure.
- Contain **valve**
- No pulses.
- Blood flows smoothly
- Small veins are called **venules**.

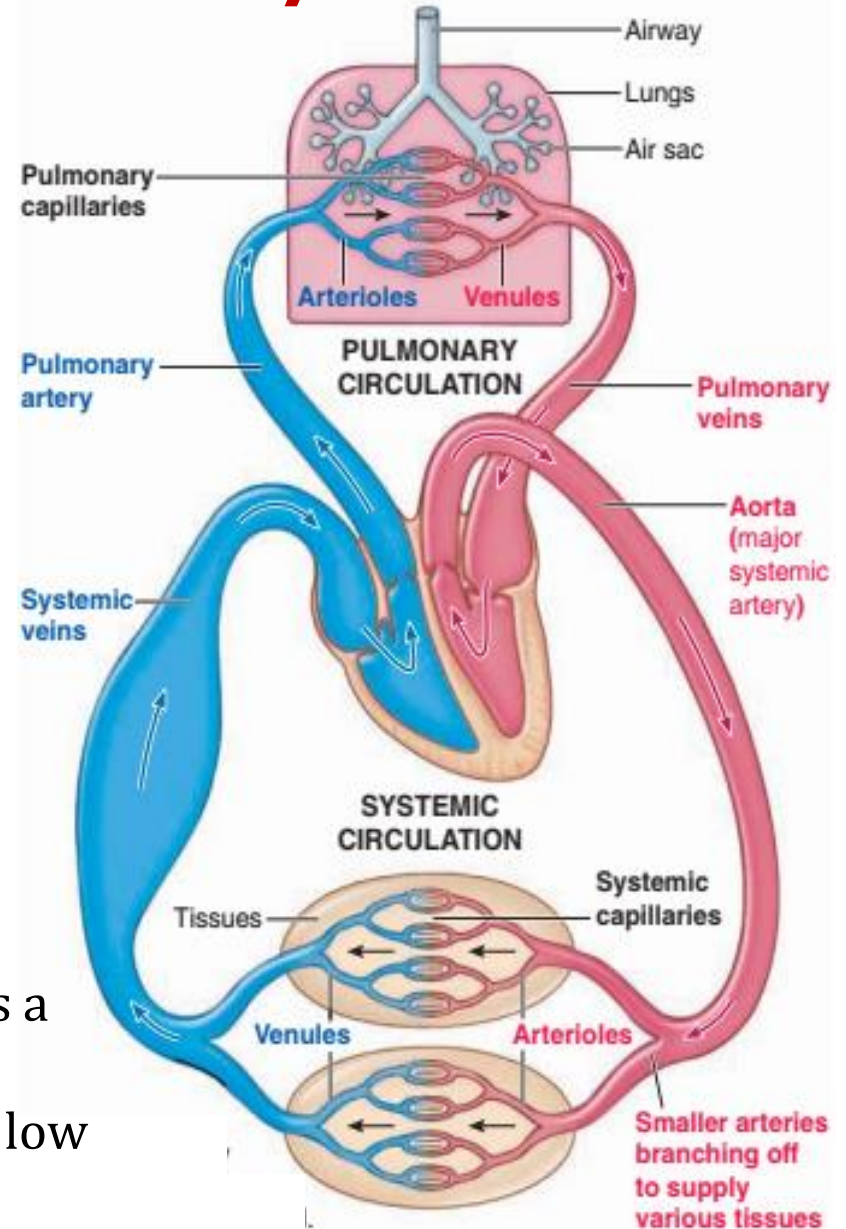
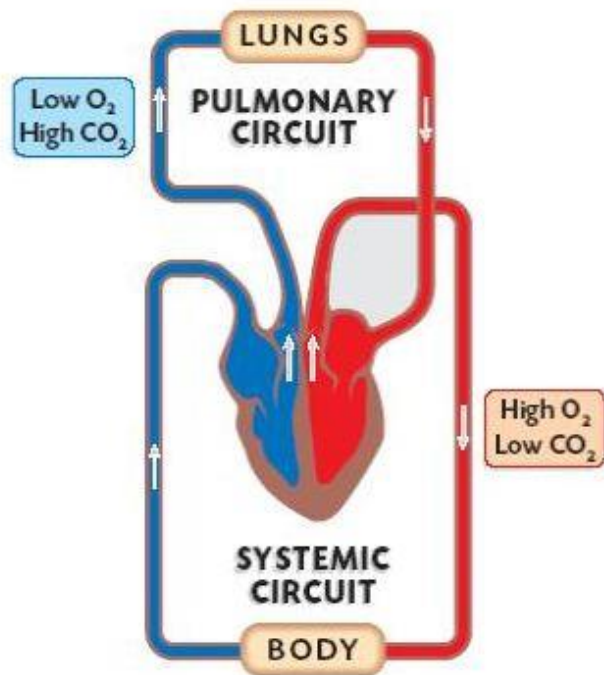
Blood Vessels - Capillaries

16

- **Capillaries** connect arteries and veins.
- They are a finely divided network of tiny tubes that exchange food, oxygen and wastes between blood and body cells.
- Capillaries leak nutrients from the bloodstream to all cells in the body.
- Cell wastes diffuse through the capillary back into the blood.



Systemic Circuit & Pulmonary Circuit

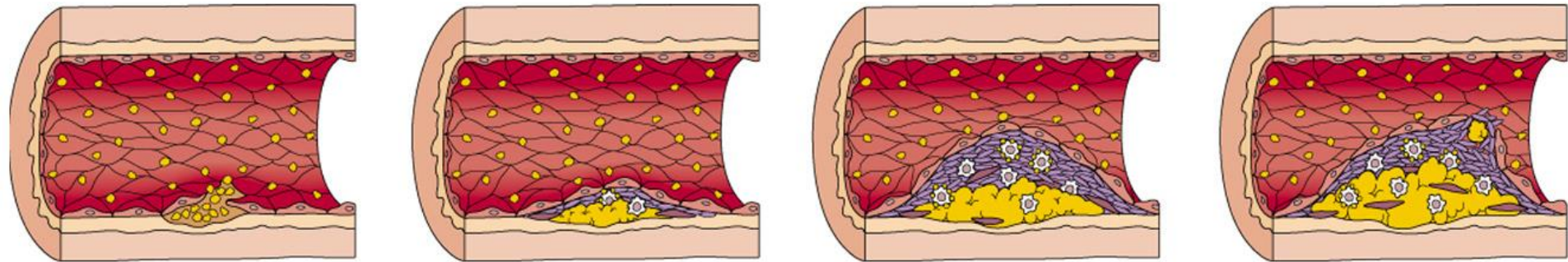


- **Oxygenated blood** – Blood that contains a high percentage of oxygen
- **Deoxygenated blood** – Blood that has a low percentage of oxygen

Disorders

□ Atherosclerosis/ Arteriosclerosis:

- deposits inside arteries (**plaques** - made up of fat, cholesterol, calcium, and other substances found in the blood)
- develop in inner wall of the arteries, narrowing their channel
- increase blood pressure
- increase risk of heart attack, stroke, kidney damage



Plaque buildup begins when endothelial cells lining the arteries are damaged by smoking, high blood pressure, oxidized LDL, and other causes; excess cholesterol particles collect beneath these cells.

In response to the damage, platelets and other types of cells collect at the site; a fibrous cap forms, isolating the plaque within the artery wall. An early-stage plaque is called a fatty streak.

Chemicals released by cells in and around the plaque cause further inflammation and buildup; an advanced plaque contains LDL, white blood cells, connective tissue, smooth muscle cells, platelets, and other compounds.

The narrowed artery is vulnerable to blockage by clots. The risk of blockage and heart attack rises if the fibrous cap cracks (probably due to destructive enzymes released by white blood cells within the plaque).

Disorders

25

Heart Attack

- Acute **myocardial infarction**
- Interruption of oxygen supply to the heart
- Causes death of the heart muscle
- Leading cause of death in both men and women

Stroke

- Strokes usually result from rupture or blockage of arteries in the head
- Brain cells are starved of oxygen and nutrients
- Loss of function may occur
- Can cause paralysis, loss of ability to speak or death.

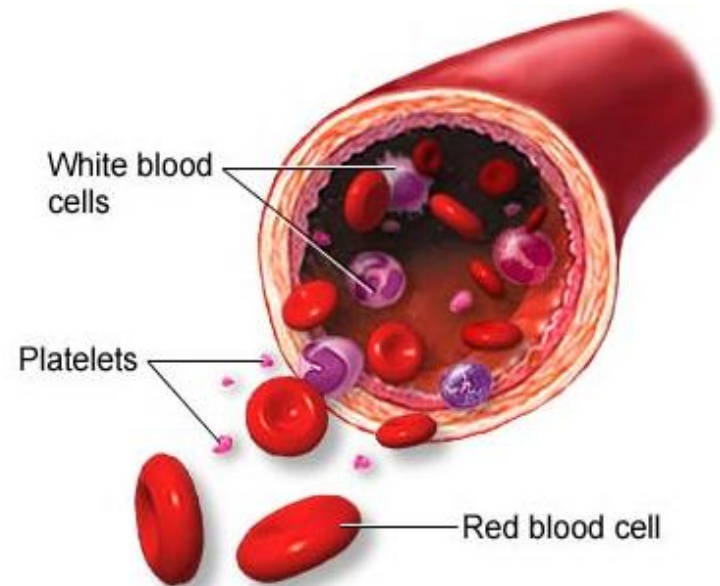
What is Blood?

30

Blood is a specialized liquid connective tissue that circulates through the arteries and veins, carrying essential substances throughout the body. It consists of several components:

1. Plasma
2. Blood cells

Blood cells are produced in bone marrow from stem cells.



Composition of Blood

31

1. Plasma

- ▣ Fluid part of blood [soluble portion]
- ▣ Carries carbon dioxide, hormones and waste

2. Blood cells

a) Red blood cells

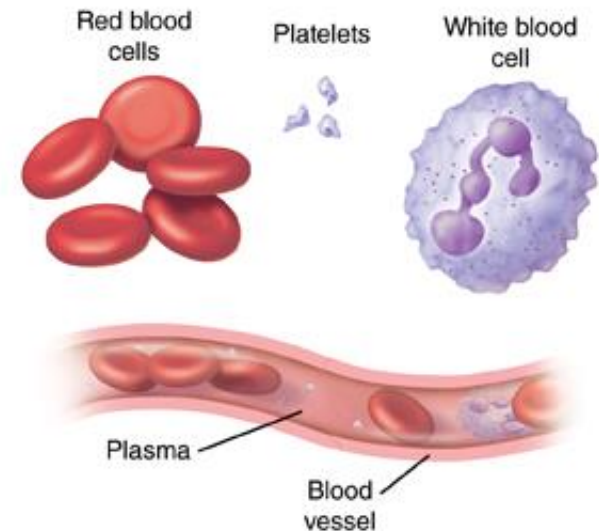
- ▣ Contain haemoglobin which carries oxygen

b) White blood cells

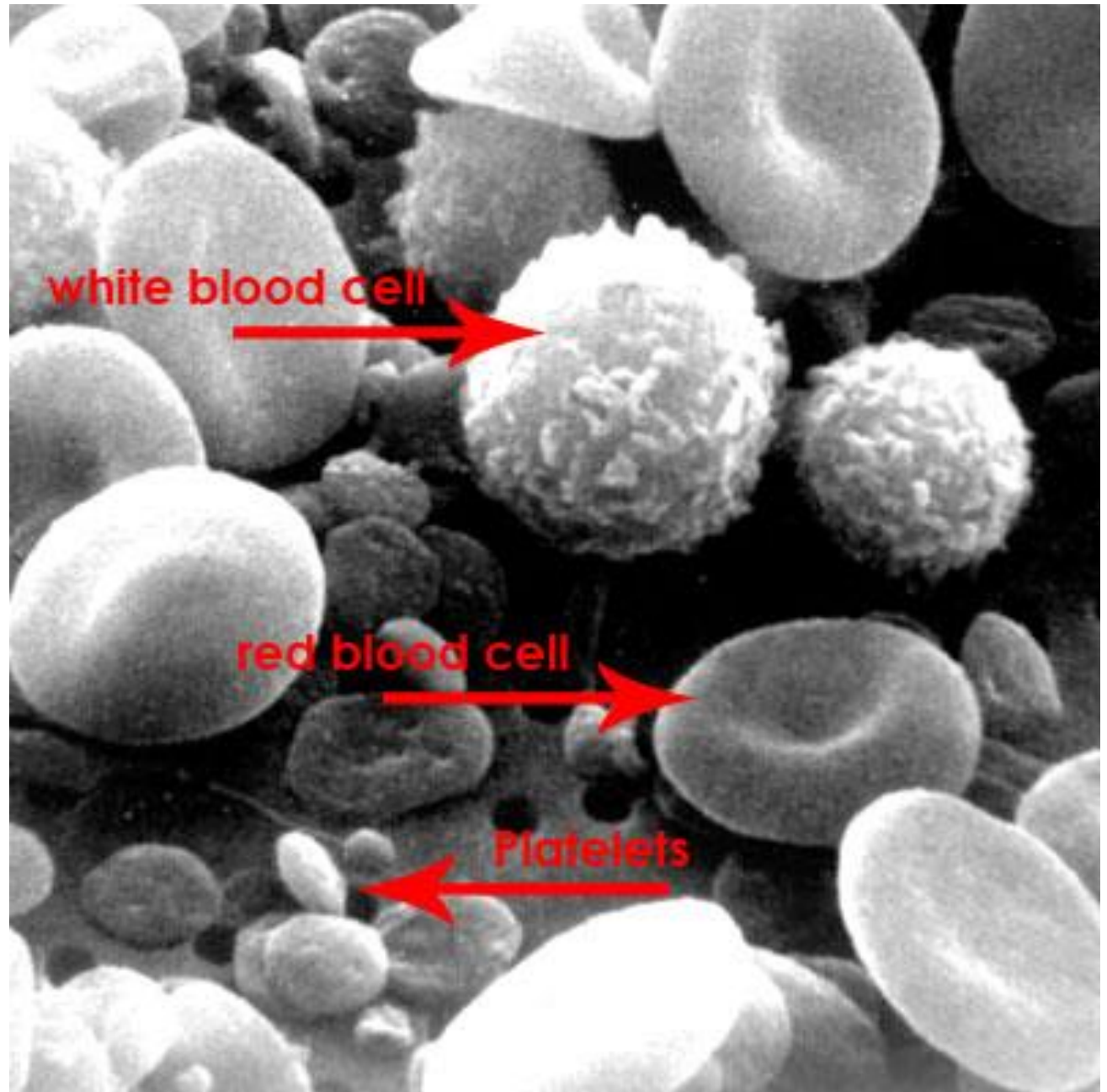
- ▣ An important part of the immune system, they produce antibodies and destroy harmful microorganisms

c). Platelets

- ▣ Clump together to form clots
- ▣ Protect the body by stopping bleeding



**Figure – RBC,
WBC and
Platelets**



Function of Blood

Transport: as delivery and waste removal system:

- **Oxygen** from lungs → tissues (via hemoglobin in RBCs)
- **Carbon dioxide** from tissues → lungs for exhalation
- **Nutrients** (glucose, amino acids, lipids) from digestive tract → cells
- **Hormones** from endocrine glands → target organs
- **Waste products** (urea, creatinine) → kidneys for excretion

Regulation: keep internal environment stable (**homeostasis**):

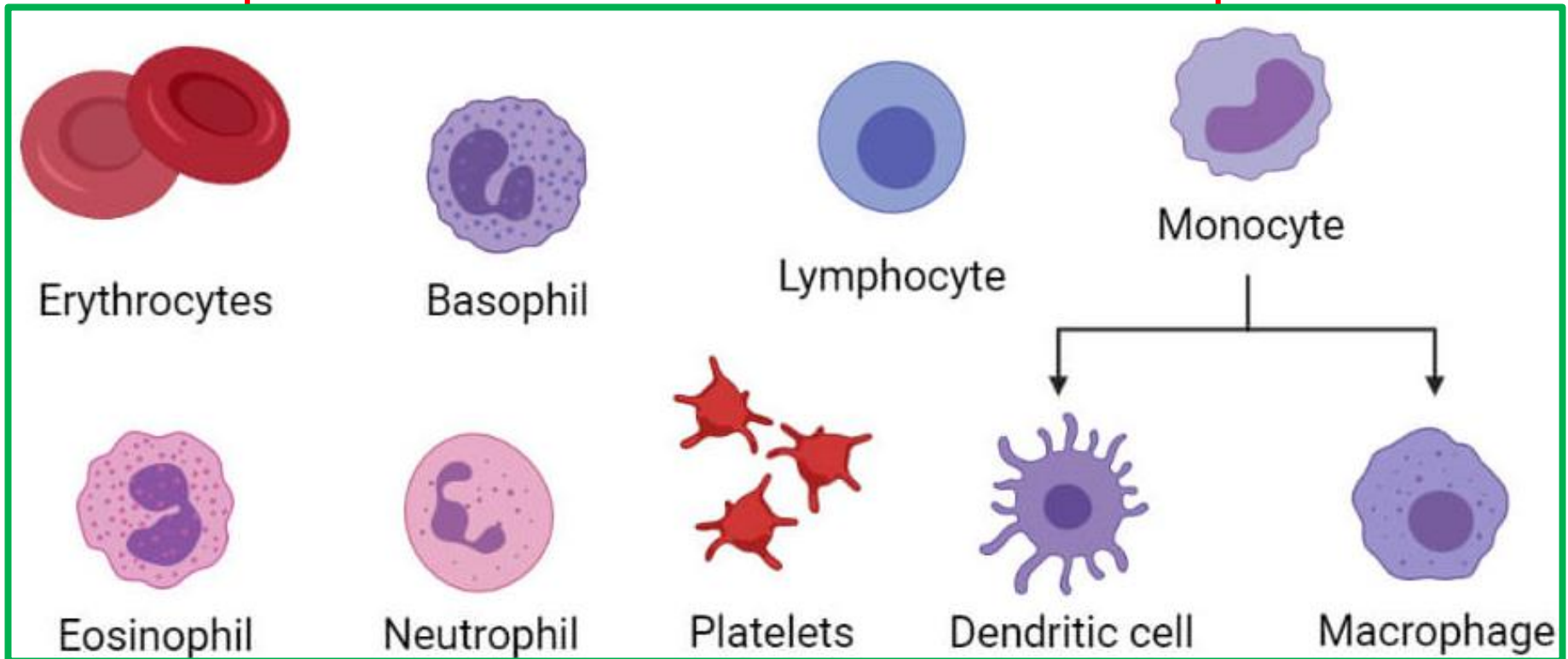
- **pH balance** → bicarbonate, proteins in plasma keep pH ~7.45
- **Temperature regulation** → absorbs & distributes heat
- **Fluid balance** → plasma proteins (albumin) maintain osmotic pressure

Protection: Blood plays a key role in defense and repair:

- **Immunity** → White blood cells and antibodies fight infections
- **Clotting** → Platelets and clotting factors prevent injurious blood loss
- **Detoxification** → Transports toxins to liver/kidneys for neutralization & excretion

COMPONENTS OF BLOOD

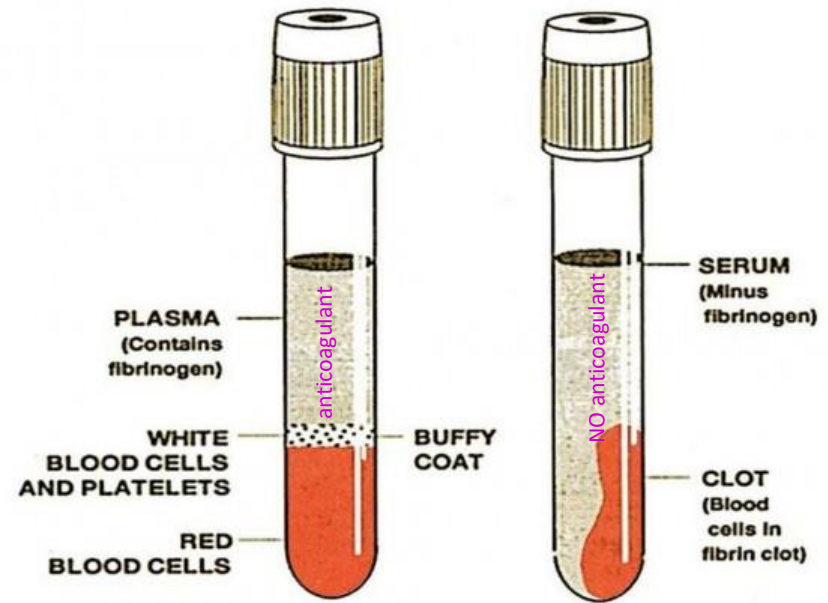
Blood



Difference between Plasma and Serum

Plasma is the liquid component of blood that remains **after cells** (red blood cells, white blood cells, and platelets) are **removed by centrifugation** and clotting factors (fibrinogen) are allowed to clot.

Serum is the liquid portion of blood that remains **after clotting factors, primarily fibrinogen, are removed** from plasma during blood clotting (essentially plasma *without fibrinogen and other clotting factors*).



Types Of Blood

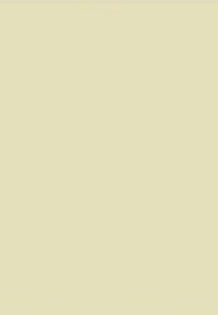
Human blood is grouped into four types:

A, B, AB, and O based on two antigens A and B

Each blood type is also grouped by its Rhesus factor, or **Rh factor**:

Rh+individual has Rh antigen

Rh-.....individual do not has Rh antigen;

	Type A	Type B	Type AB	Type O
Red blood cells	Antigen A 	Antigen B 	Antigens A and B 	Neither A nor B antigens 
Plasma antibodies	 B	 A	Neither A nor B 	 A and B

Universal Blood donor and recipient

51

- Type **O⁻ ve blood** is considered the “**universal donor**” because it can be donated to people of any blood type. It contains no antigen.
- Type **AB⁺ ve blood** is considered the “**universal recipient**” because people with this type can receive any blood type. It contain both the antigens and will not produce any antibody against foreign antigens.

RBC surface antigen and plasma-antibody

	Group A	Group B	Group AB	Group O
Red Blood Cell Type	